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Experiment: Travelling Salesman Problem (TSP)

Aim

To implement the **Travelling Salesman Problem (TSP)** using Python to find the minimum cost path that visits every city exactly once and returns to the starting city.

Objectives

- To understand the Travelling Salesman Problem (TSP).
- To implement TSP using Python.
- To find the shortest possible route that visits all cities exactly once.
- To calculate the minimum travel cost.
- To understand the application of optimization techniques in Artificial Intelligence.

Algorithm

1. Represent the distances between cities using a distance matrix.
2. Select the starting city.
3. Generate all possible routes (permutations) of the remaining cities.
4. Calculate the total cost of each route, including the return to the starting city.
5. Compare the costs of all routes.
6. Select the route with the minimum total cost.
7. Display the optimal route and its minimum cost.

Python Program

```
from itertools import permutations
```

```
# Distance matrix
```

```
graph = [
```

```
    [0, 10, 15, 20],
```

```
[10, 0, 35, 25],  
[15, 35, 0, 30],  
[20, 25, 30, 0]  
]
```

```
n = len(graph)  
cities = range(n)
```

```
min_cost = float('inf')  
best_path = ()
```

```
# Generate all possible routes  
for path in permutations(cities[1:]):  
    current_path = (0,) + path + (0,)  
    cost = 0  
  
    for i in range(len(current_path) - 1):  
        cost += graph[current_path[i]][current_path[i + 1]]  
  
    if cost < min_cost:  
        min_cost = cost  
        best_path = current_path
```

```
print("Optimal Path:", best_path)  
print("Minimum Cost:", min_cost)
```

Sample Output

Optimal Path: (0, 1, 3, 2, 0)

Minimum Cost: 80

Out Put

```
5     [0, 10, 15, 20],
6     [10, 0, 35, 25],
7     [15, 35, 0, 30],
8     [20, 25, 30, 0]
9 ]
10
11 n = len(graph)
12 cities = range(n)
13
14 min_cost = float('inf')
15 best_path = ()
16
17 # Generate all possible routes
18 for path in permutations(cities[1:]):
19     current_path = (0,) + path + (0,)
20     cost = 0
21
22     for i in range(len(current_path) - 1):
23         cost += graph[current_path[i]][current_path[i + 1]]
24
25     if cost < min_cost:
26         min_cost = cost
27         best_path = current_path
28
29 print("Optimal Path:", best_path)
30 print("Minimum Cost:", min_cost)
```

Optimal Path: (0, 1, 3, 2, 0)
Minimum Cost: 80

=== Code Execution Successful ===

Result

The Travelling Salesman Problem was successfully implemented using Python. The program generated all possible routes and identified the route with the minimum travel cost.

Conclusion

The Travelling Salesman Problem (TSP) was solved using the **brute-force approach** by evaluating all possible routes and selecting the one with the minimum cost. Although this method guarantees the optimal solution, it becomes computationally expensive as the number of cities increases. The experiment demonstrates the importance of optimization techniques in Artificial Intelligence and route planning applications.